1. A particle of mass $m = 2\text{kg}$ is moving along a straight line and is driven by a force varying with time: $F = 4 - t$, where the force F is in newton and the time t is in second. At t=0, the particle is at x=0, and the velocity is $v_0 = 2\,\text{m/s}$, along the positive direction.
(a) Determine the maximum positive velocity of the particle.
(b) Determine the motional function of the particle.
(c) Determine the time that the particle starts to move backward.
(d) Determine the work done on the particle when it comes back to x=0.
[Solution] (a) Acceleration: $a = F/m = 2 - t/2$; velocity: $v = 2 + 2t - t^2/4$;
Maximum velocity: $\mathrm{d}v/\mathrm{d}t = a = 0; t = 4; v_{\max} = 6$ (m/s).
(b) $v = 2 + 2t - t^2/4, x = 2t + t^2 - t^3/12$ (m).
(c) $v = 0, t = 4 + 2\sqrt{6}$ (s).
(d) $x = 0, t = 6 + 2\sqrt{15}, W = mv^2/2 - mv_0^2/2 = 156 + 40\sqrt{15}$ (J).

2. A particle is in a potential field defined on the x axis:

$$U = x\mathrm{e}^{-x^2}$$

(a) Find the positions of all the equilibria, and determine their stabilities.
(b) Determine the maximum force exerted on the particle, and the corresponding position.
[Solution] (a) Equilibria: $\mathrm{d}U/\mathrm{d}x = (1 - 2x^2)\mathrm{e}^{-x^2} = 0, x = \pm 1/\sqrt{2}$.
$\mathrm{d}^2U/\mathrm{d}x^2 = (4x^3 - 6x)\mathrm{e}^{-x^2}$; for $x = 1/\sqrt{2}$, $\mathrm{d}^2U/\mathrm{d}x^2 < 0$, unstable; for $x = -1/\sqrt{2}$, $\mathrm{d}^2U/\mathrm{d}x^2 > 0$, stable.
(b) $F = -\mathrm{d}U/\mathrm{d}x$. The extrema: $\mathrm{d}F/\mathrm{d}x = -\mathrm{d}^2U/\mathrm{d}x^2 = 0$, $x = 0$ or $\pm\sqrt{6}/2$, $F = 1$ or $F = -2\mathrm{e}^{-3/2}$; the maximum: $F = 1, x = 0$.

3. A car of mass m starts to move from x=0 under a driving power P, which is inversely proportional to its speed:

$$P = \alpha/v$$

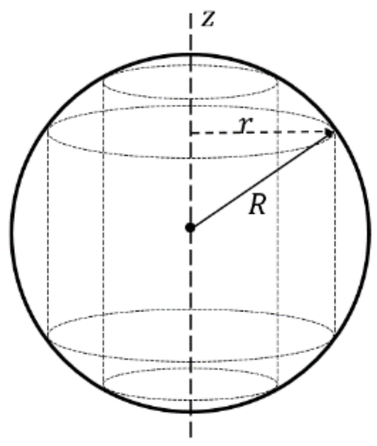
If there is no energy dissipation (损耗), determine its motional function. (**Hint:** make use of the formula P=Fv, establish a differential equation and solve it.)
[Solution] $F = P/v = m\,\mathrm{d}v/\mathrm{d}t$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{P}{mv} = \frac{\alpha}{mv^2} \rightarrow v^2\mathrm{d}v = \frac{\alpha}{m}\mathrm{d}t \rightarrow \frac{1}{3}v^3 = \frac{\alpha}{m}t + C$$

Initial condition: t=0, v=0, so C=0.

$$v = \left(\frac{3\alpha}{m}t\right)^{1/3}, x = \frac{3}{4}\left(\frac{3\alpha}{m}\right)^{1/3} t^{4/3}$$

4. A uniform solid sphere of mass $m = 10\text{kg}$ and radius of $R = 1\text{m}$ can rotate around the z-axis, which goes through its center.
(a) Determine its moment of inertia. (**Hint:** in this case of division, the height of the "thin tube" varies with the radius. You may use some integral formula.)
(b) If there is a force $\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$, exerted on it, in newton, and the position is $0.8\mathbf{j} - 0.6\mathbf{k}$, in meter, determine its angular acceleration. (**Attention:** for fixed-axis rotation, the component of the torque perpendicular to the axis will be canceled.)



[Solution] (a) Divide the sphere into thin tubes of radius r:

$$h = 2\sqrt{R^2 - r^2}, \mathrm{d}V = 2\pi rh\mathrm{d}r = 4\pi r\sqrt{R^2 - r^2}\mathrm{d}r$$

$$\mathrm{d}I = r^2\mathrm{d}m = r^2\rho\mathrm{d}V = 4\pi\rho r^3\sqrt{R^2 - r^2}\mathrm{d}r$$

If we set $r = R\sin\theta$, then $\mathrm{d}I = 4\pi\rho R^4\sin^3\theta\cos\theta\,\mathrm{d}(R\sin\theta) = 4\pi\rho R^5\sin^3\theta\mathrm{d}(\sin^2\theta)$

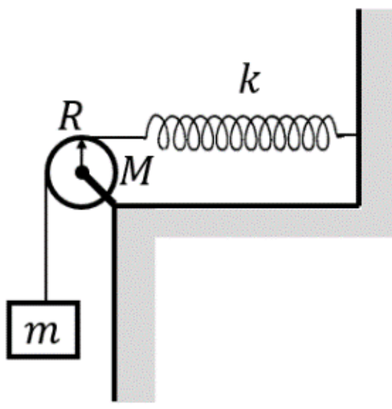
$$4\sin^3\theta\cos^2\theta = \sin^2 2\theta\sin\theta = \frac{1}{2}(1 - \cos 4\theta)\sin\theta = \frac{1}{2}\sin\theta + \frac{1}{4}\sin 3\theta - \frac{1}{4}\sin 5\theta$$

$$I = \pi\rho R^5\int_0^{\pi/2}\left(\frac{1}{2}\sin\theta + \frac{1}{4}\sin 3\theta - \frac{1}{4}\sin 5\theta\right)\mathrm{d}\theta = \frac{8}{15}\pi\rho R^5$$

$$m = \rho\frac{4}{3}\pi R^3, I = \frac{2}{5}mR^2 = 4(\text{kg}\cdot\text{m}^2)$$

(b) $\boldsymbol{\tau} = \mathbf{r}\times\mathbf{F} = -1.8\mathbf{i} - 1.2\mathbf{j} - 1.6\mathbf{k}$, the effective component is $-1.6\mathbf{k}$.

5. As is shown, a spring of stiffness k is tied to the wall, and the other end is connected to a weight of mass m with a string passing over a cylindrical pulley (圆柱形滑轮) of mass M and radius R. There is no slipping (打滑) between the pulley and the string. The mass of the spring and the string, and all the resistance are neglected. Assume that the weight cannot swing horizontally.



(a) How many degrees of freedom does the system have? (**Attention**: "no slipping" indicates the speeds of the string must be equal to that of the edge of the pulley, and it is a constraint.)

(b) Give out the Lagrangian function of the system. Make the expression of potential energy reasonable according to the general coordinate. (**Hint**: the moment of inertia of a uniform cylinder to its axis is $I = \frac{1}{2}MR^2$)

(c) Give out the Hamiltonian canonical equations.

[Solution] (a) The spring: N=1; the fixed-axis pulley: N=1; the weight: N=1 (because it cannot swing horizontally.); the string: l=1 (the stretch of the spring should always be sync with the displacement weight because of the connection); no slipping: l=1.

Finally, there is only one degree of freedom.

(b) Potential energy: set the position where the spring is relaxed as x=0, and assume that x increases when the weight descends, then $U = \frac{1}{2}kx^2 - mgx$.

Kinetic energy: $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$; for no slipping: $v = R\omega$, hence

$$K = \frac{1}{2}mv^2 + \frac{1}{4}Mv^2 = \left(\frac{m}{2} + \frac{M}{4}\right)\dot{x}^2, L = K - U = \left(\frac{m}{2} + \frac{M}{4}\right)\dot{x}^2 - \frac{1}{2}kx^2 + mgx$$

(c) The general momentum and the Hamiltonian function:

$$p = \frac{\partial L}{\partial \dot{x}} = \left(m + \frac{M}{2}\right)\dot{x}, H = p\dot{x} - L = \left(\frac{m}{2} + \frac{M}{4}\right)\dot{x}^2 + \frac{1}{2}kx^2 - mgx = K + U$$

The canonical equations:

$$\dot{p} = -\frac{\partial H}{\partial x} \rightarrow \left(m + \frac{M}{2}\right)\ddot{x} = mg - kx$$

$$\dot{x} = \frac{\partial H}{\partial p} = \dot{x}$$